

# ISO New England

*Analysis of Energy Generation and the Pathway to Electrification*

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## Introduction

ISO New England (ISO-NE) oversees the electric power system for Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont. The regional grid serves approximately 15.1 million people, representing about 4% of the U.S. population, and supplies electricity to roughly 7.6 million retail customers (*ISO New England, 2023*). ISO-NE is interconnected with neighboring system operators, including the New York Independent System Operator (NYISO), Énergie New Brunswick Power, and Hydro-Québec TransÉnergie, allowing for the exchange of electricity across regional boundaries. Historically, peak electric demand in the region has occurred during the summer months due to high air-conditioning loads.

As the region transitions toward electrification of transportation, heating, and industry, the structure and operation of the power system are expected to evolve significantly. Increasing reliance on electricity, coupled with the need to reduce carbon emissions, introduces new challenges related to resource adequacy, system flexibility, and peak demand management. Understanding the current composition and behavior of the ISO-NE grid is therefore critical for evaluating future system requirements.

This report presents a quantitative analysis of electricity generation and consumption in ISO New England using data from ISO-NE and the U.S. Energy Information Administration (EIA). The analysis examines regional demand patterns, generation sources, and system structure to identify key constraints and considerations that will influence the region's pathway toward a more electrified and sustainable energy system.

# Electricity Generation and Consumption in ISO-NE

## ISO New England: Regional Electricity Statistics

ISO New England provides comprehensive data on regional electricity demand, generation, and system reliability. In 2025, ISO-NE reported a total annual energy served of 117,744 GWh. For historical comparison, the region's all-time highest annual energy served was 136,355 GWh in 2005, indicating that present annual electricity use remains below its historical maximum despite continued economic activity and electrification pressures (*ISO New England, 2023*).

Peak demand is especially important because the power system must be sized to meet the highest instantaneous load, not simply total annual consumption. ISO-NE's all-time summer peak demand was 28,130 MW on August 2, 2006, while its all-time winter peak was 22,818 MW on January 15, 2004 (*ISO New England, 2023*). In the current report dataset, the 2025 summer and winter peak values are listed as 24,978 MW and 20,483 MW, respectively, showing that recent peak demand remains below historical records but still requires substantial dispatchable capacity and system flexibility (*ISO New England, 2026*).

| 2025 Net Energy and Peak Demands |             |
|----------------------------------|-------------|
| Net energy for load              | 117,744 GWh |
| Summer peak demand               | 24,978 MW   |
| Winter peak demand               | 20,483 MW   |

Table 1: Tabulated Net Energy and Peak Demands (*ISO New England, 2026*).

The state-level breakdown of electricity generation and consumption (below) within ISO New England provides important insight into the regional structure of the power system and the flow of energy across the grid. While ISO-NE operates as an integrated system, individual states exhibit significant imbalances between in-state generation and electricity demand. For example, states such as Massachusetts and Vermont consume substantially more electricity than they generate, while others like New Hampshire and Connecticut are net exporters of electricity. These disparities highlight the critical role of transmission infrastructure and inter-state power flows in maintaining system reliability. Additionally, this data reveals geographic dependencies that are essential for understanding how the region currently balances supply and demand, and it

provides a foundation for evaluating how future electrification and generation expansion must be distributed spatially to ensure a resilient and efficient grid.

| State         | Generation<br>(GWh) | Net Energy<br>for Load<br>(GWh) | Net Balance<br>(GWh) | Primary Source |
|---------------|---------------------|---------------------------------|----------------------|----------------|
| Connecticut   | 43,649              | 27,216                          | -16,433              | Natural Gas    |
| Massachusetts | 23,413              | 54,044                          | 30,631               | Natural Gas    |
| New Hampshire | 17,379              | 11,614                          | -5,765               | Nuclear        |
| Maine         | 13,236              | 11,239                          | -1,997               | Natural Gas    |
| Rhode Island  | 9,775               | 7,793                           | -1,982               | Natural Gas    |
| Vermont       | 1,147               | 4,891                           | 3,744                | Hydroelectric  |

Table 2: Net Balance per State (*ISO New England, 2025*)

## Electricity Generation Mix

ISO New England's electricity generation mix is dominated by natural gas, followed by nuclear energy, hydropower, solar, wind, and other renewable or waste-based resources. In 2025, natural gas supplied the largest share of in-region generation, producing approximately 60,446 GWh, or about 54% of total supply. Nuclear generation was the second-largest source, producing approximately 27,620 GWh, or about 25% of total supply. Hydropower, solar, and wind contributed smaller but important shares of the generation mix (ISO New England, 2026).

This generation profile reflects ISO-NE's current balance between reliability, flexibility, and decarbonization. Nuclear power provides steady baseload generation, while natural gas serves as the primary dispatchable resource that can ramp output to follow changes in demand. This makes natural gas valuable for reliability, especially during peak periods, but it also creates continued dependence on fossil fuels and exposes the region to fuel-price volatility and winter fuel-supply constraints.

Over recent years, New England has decommissioned its coal and oil fired power stations and in many instances, replaced them with natural gas power stations. There are 126 utility-scale natural gas power stations operating in the region. While some nuclear stations have been shut down, two remain: Millstone Power Station (CT) and Seabrook Station (NH). Millstone Power

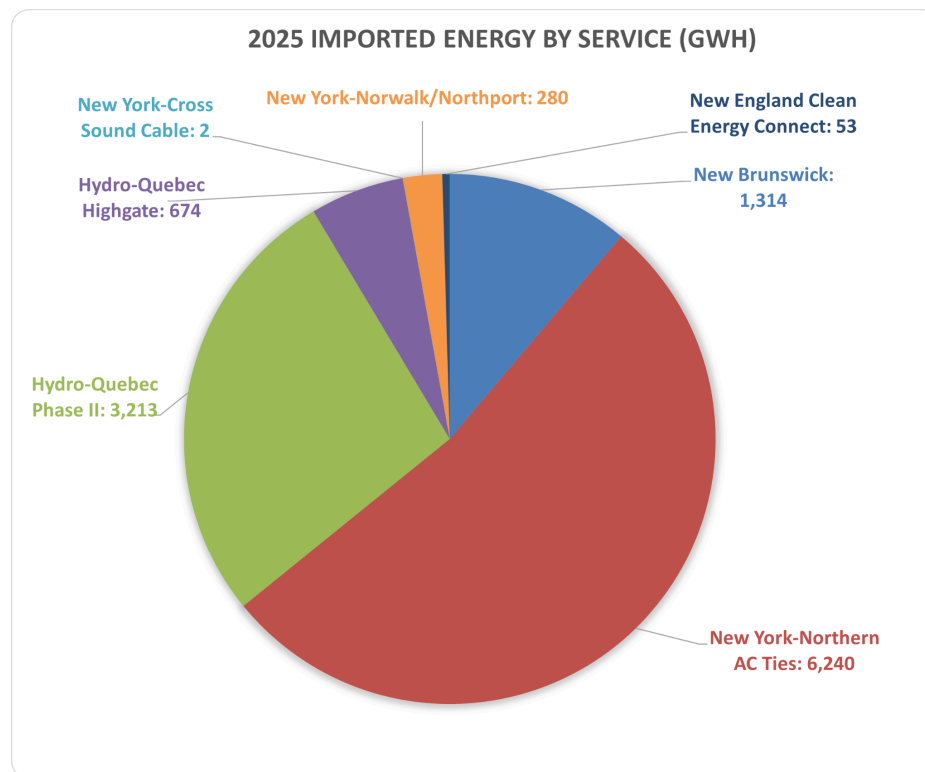
station opened in 1970 and is permitted to operate through 2045, while life-extension plans are underway to keep it in operation until 2065. Seabrook station started operating in 1990 and is permitted to operate until March 2050. As with most nuclear power stations in the US, permit cycles continue to be renewed to extend operation past the original planned life. Also of note, is the contribution of approximately 85 hydroelectric generation facilities in Vermont.

Renewable resources such as wind and solar are becoming increasingly important as ISO-NE transitions toward a lower-carbon electricity system. However, their output varies with weather and time of day, meaning they cannot fully replace dispatchable generation without additional grid flexibility, storage, transmission expansion, or demand-response resources. As a result, the current generation mix shows both progress toward decarbonization and the remaining challenge of maintaining reliability as fossil generation is reduced.

| Source                    | Energy Generation (GWh) | % of Generation |
|---------------------------|-------------------------|-----------------|
| Natural Gas               | 60,446                  | 54.14%          |
| Nuclear                   | 27,620                  | 24.74%          |
| Hydro                     | 7,064                   | 6.33%           |
| Solar                     | 4,840                   | 4.34%           |
| Wind                      | 4,622                   | 4.14%           |
| Refuse                    | 2,556                   | 2.29%           |
| Wood                      | 2,012                   | 1.80%           |
| Oil                       | 1,147                   | 1.03%           |
| Other                     | 705                     | 0.63%           |
| Landfill Gas              | 339                     | 0.30%           |
| Coal                      | 268                     | 0.24%           |
| Methane                   | 21                      | 0.02%           |
| Price-Responsive Demand   | 10                      | 0.01%           |
| <b>Total Supply (GWh)</b> | <b>111,649</b>          | <b>100%</b>     |

Table 3: 2025 Generation Profile for ISO-NE (*ISO New England, 2026a*).

Around 8-10% of the annual energy needs are imported from New York Independent System Operator (NYISO), the Énergie New Brunswick Power, and Hydro-Québec TransÉnergie. Energy generation in NYISO consists largely of natural gas followed by hydroelectric. The majority of the New Brunswick generation is nuclear-based, followed by hydroelectric. Quebec generation is hydroelectric-based. The hydropower from Canada is both dispatchable (flexible) and low-carbon. It importantly helps with addressing peak demand issues and stems intermittency of renewable energy (wind and solar). Imports are critical for ISO-NE, as they promote reliability, price stability, and assist in efforts toward decarbonization (*ISO New England, 2026a*). The figure below depicts the services through which energy was imported in 2025.



*Figure 1: 2025 Imports by Service (ISO New England, 2026c)*

The graph below depicts the portion of Net Energy Load that is generated within the region. The red region is representative of the electricity that is imported each year. This quantity varies from year to year, but is generally around 8-10% of the NEL.



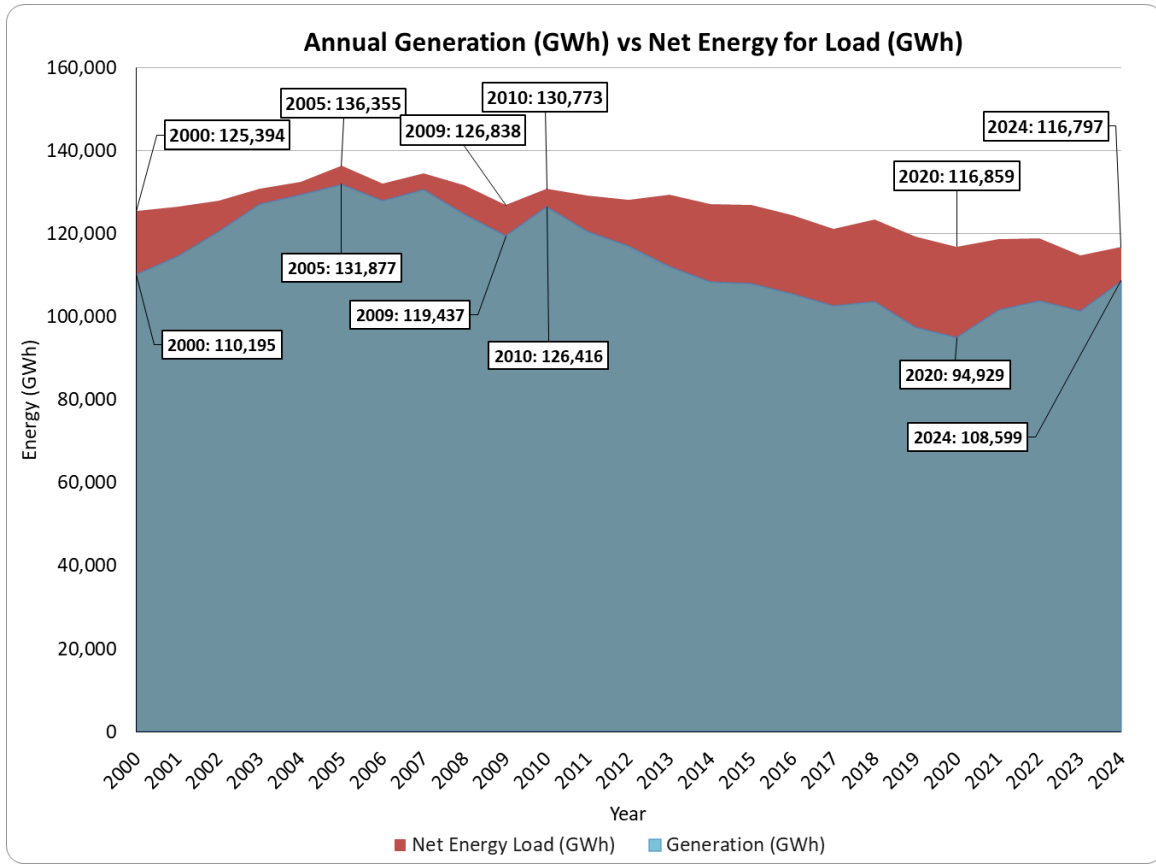


Figure 2: Annual Generation vs. Net Energy for Load (ISO New England, 2025)

## Real-Time Grid Operations from the US Energy Information Administration (EIA)

The U.S. Energy Information Administration (EIA) Real-Time Operating Grid provides hourly data on electricity demand, generation, and power flows, offering insight into how ISO New England balances the power system in real time. Unlike long-term energy planning metrics, real-time grid operation requires that electricity supply and demand be matched continuously, as large-scale energy storage is limited and imbalances can lead to system instability.

In practice, this means that system operators must constantly adjust generation output to respond to fluctuations in demand throughout the day. Natural gas generation typically dominates real-time dispatch because it is highly flexible and can ramp output quickly to meet changing load conditions. In contrast, nuclear generation provides steady baseload power but has limited ability to adjust output, while renewable sources such as wind and solar are inherently variable and dependent on weather conditions.

As a result, real-time grid reliability depends on a combination of flexible and non-flexible resources. Dispatchable generation, imports, and hydropower help balance variability in renewable output and changes in demand, ensuring that the system remains stable across hourly and sub-hourly timescales. This highlights the importance of flexibility in the ISO-NE system and underscores the operational challenges associated with increasing shares of variable renewable energy.

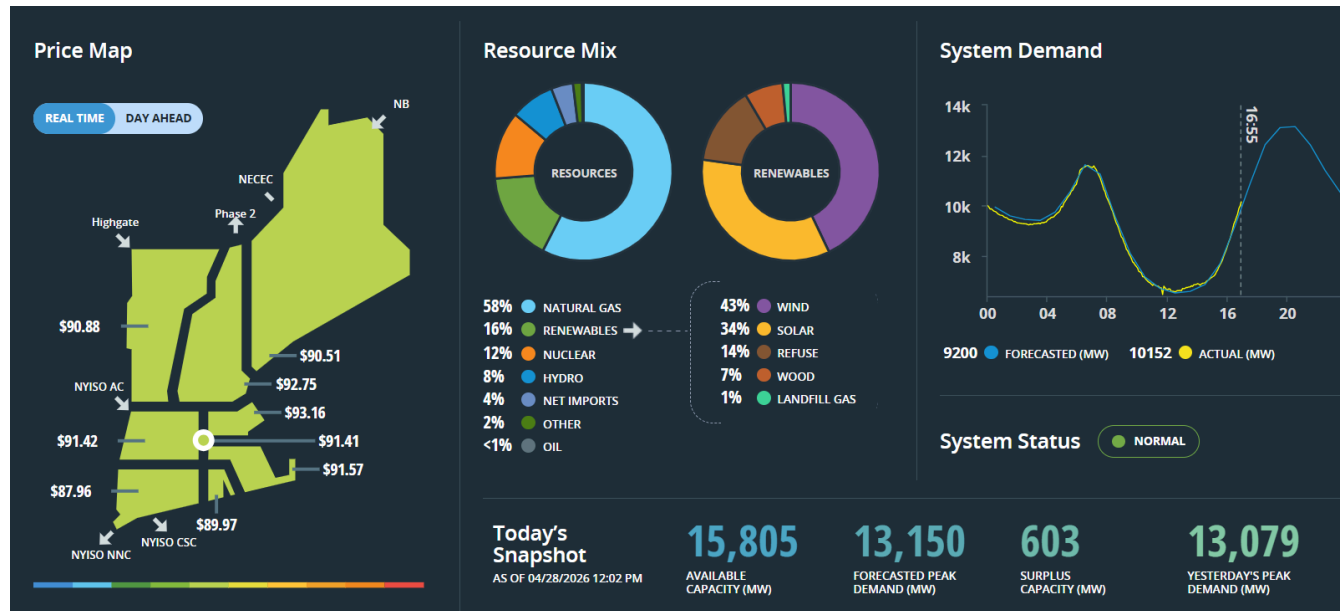


Figure 3: ISO-NE Dashboard April 28, 2026 (ISO New England, 2026c).

## Transmission and Regional Structure

ISO New England divides the region into multiple load and reserve zones to manage congestion, maintain reliability, and reflect differences in local electricity supply and demand. The eight wholesale load zones are Vermont, Maine, New Hampshire, Western/Central Massachusetts, Northeast Massachusetts, Southeast Massachusetts, Connecticut, and Rhode Island. These zones are used in the wholesale electricity market to account for local transmission constraints and regional differences in electricity pricing. ISO-NE also defines reserve zones, including Connecticut, Southwest Connecticut, Northeast Massachusetts/Boston, and Rest-of-System, to ensure that sufficient backup capacity is available in areas where the grid is more vulnerable to congestion or reliability constraints.

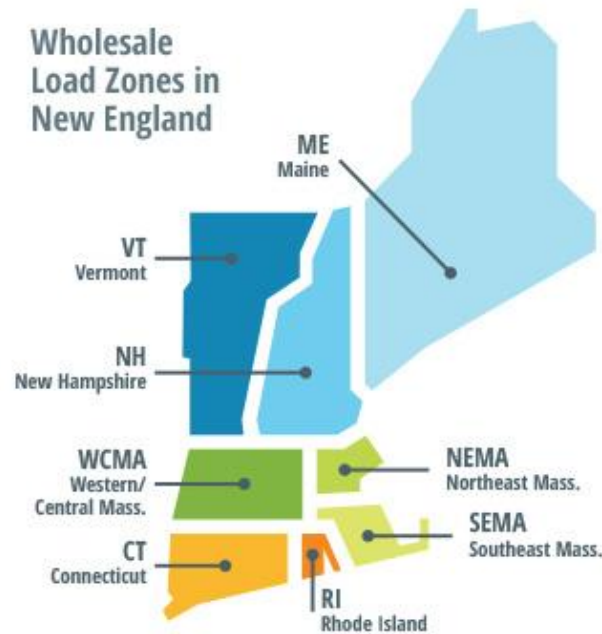


Figure 4: Wholesale Load Zones in New England (*ISO New England, n.d.-b*)

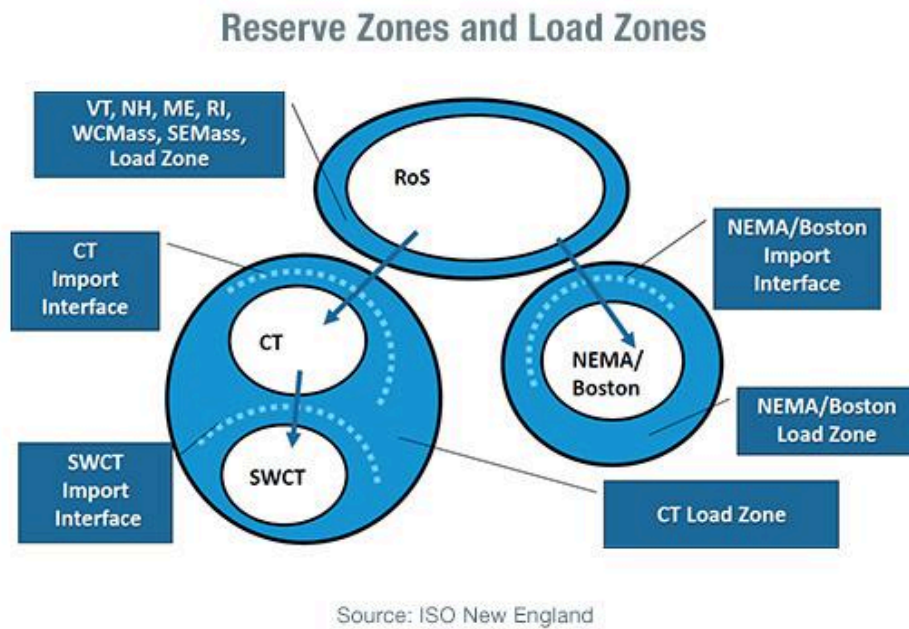


Figure 5: Reserve Zone and Load Zones (*ISO New England, n.d.-b*)

Transmission infrastructure is especially important because electricity demand and electricity generation are not evenly distributed across New England. Demand is concentrated in southern New England, particularly in Massachusetts and Connecticut, while renewable

resources, imports, and some lower-carbon generation sources are located farther north or outside the ISO-NE region. As a result, the grid must move large amounts of power across state boundaries to serve load centers reliably. ISO-NE's roughly 9,000 miles of transmission lines provide the physical network that allows generation, imports, and demand centers to operate as an integrated regional system.

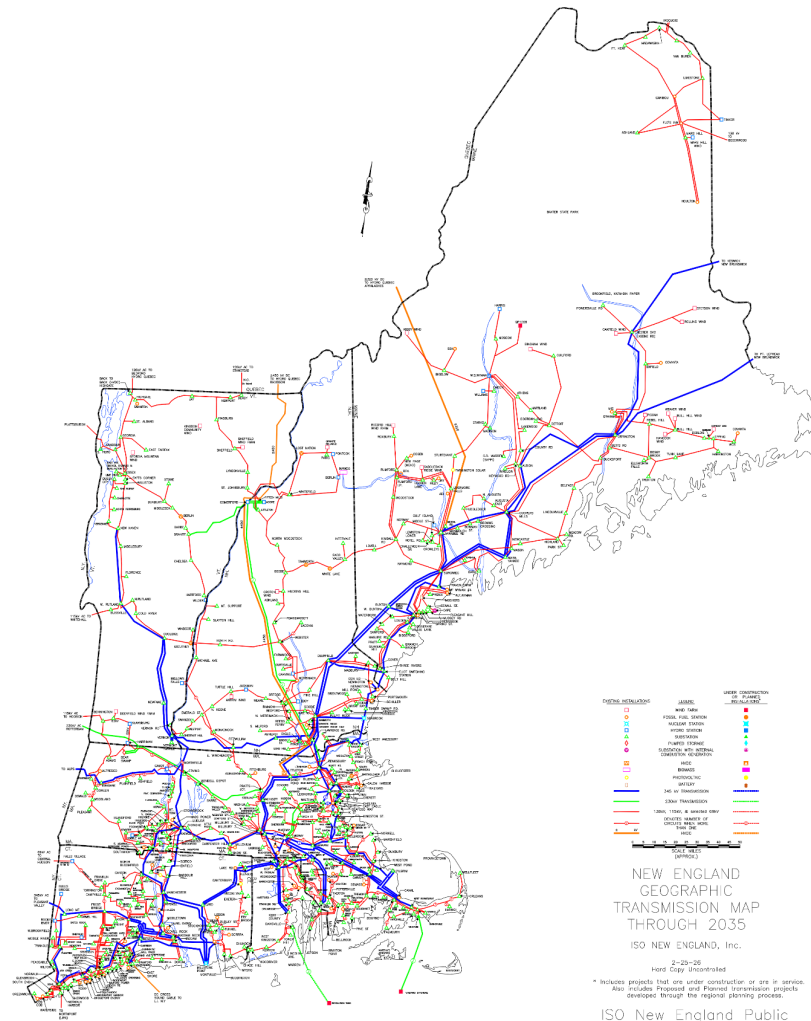


Figure 6: Regional Transmission Map (*ISO New England, 2025*)

This regional structure has important implications for future electrification. As electricity demand grows from transportation and heating electrification, transmission constraints may become more significant, especially if new renewable generation is developed far from major load centers. Therefore, transmission expansion, congestion management, and interregional coordination will be central to maintaining reliability and controlling costs in ISO-NE's future grid.

## Seasonal Demand Patterns

Historically, New England has been a summer-peaking region, as hot and humid conditions drive high air-conditioning demand. However, ISO-NE forecasts indicate that this pattern is shifting as electrification of heating and transportation increases winter electricity use. In ISO-NE's March 2026 final draft forecast, the summer net 50/50 peak is projected at 24,978 MW in 2026 and 26,515 MW in 2035, while the winter net 50/50 peak is projected at 20,483 MW in 2026 and 26,382 MW in 2035 (*Rojo, 2026*). A net 50/50 peak represents the expected peak demand under normal weather conditions, with an approximately equal chance that actual peak demand will be higher or lower. The term “net” indicates that the forecast accounts for demand reductions such as energy efficiency and behind-the-meter solar. These projections show that winter peak demand is rapidly approaching summer levels and may soon become the dominant system constraint.

This shift has important implications for system planning. Winter peaks are typically more challenging to manage because they coincide with increased demand for heating, potential constraints in natural gas supply, and reduced output from solar generation. As a result, the transition toward a winter-peaking system will require greater emphasis on fuel security, system flexibility, and resource adequacy during cold-weather conditions.

## Diurnal Demand Patterns

On a daily timescale, electricity demand follows a predictable cycle. Demand is lowest overnight, increases in the morning as residential and commercial activity begins, remains elevated throughout the day, and typically peaks in the late afternoon or early evening. ISO-NE's Three-Day System Demand Forecast and ISO Express real-time tools capture this hourly variation in demand across the region.

The increasing penetration of behind-the-meter solar generation has begun to reshape this daily load profile. Solar generation reduces midday grid demand, while demand rises sharply in the evening as solar output declines, creating a steeper ramp in net load. This pattern, often referred to as the “duck curve,” introduces additional operational challenges by requiring fast-ramping, flexible generation to meet rapid changes in demand over short time periods.

Together, these seasonal and diurnal demand patterns highlight the importance of both capacity and flexibility in the ISO-NE system. The grid must not only have sufficient generation to meet peak demand, but also the ability to respond dynamically to changes in demand across both daily and seasonal timescales.

## Peak Demand

In recent times, ISO-NE has experienced summer peaks, with demand rising the most on the hottest and most humid days due to air conditioning demand. The highest total energy served year was back in 2005, where the ISO delivered 136,355 GWh to customers. The winter prior, on January 15, 2004, the region experienced all-time high winter peak demand of 22,818 MW. The following summer, on August 2, 2006, the region experienced all-time high summer peak demand of 28,130 MW (*ISO New England, 2023*). It's clear that ISO-NE is already on the path to decarbonization by prioritizing load reduction. The northeast region saw steady growth in net energy use from 1995 to 2005 due to economic growth and the increased use of air conditioning. The 2025 equivalents reflect approximately 15% reduction in energy use. This decrease in grid demand is attributed to government incentives in energy efficiency and behind-the-meter photovoltaics. The figure below depicts the decrease in both annual energy use and peak demand experienced from 2000 to 2025.

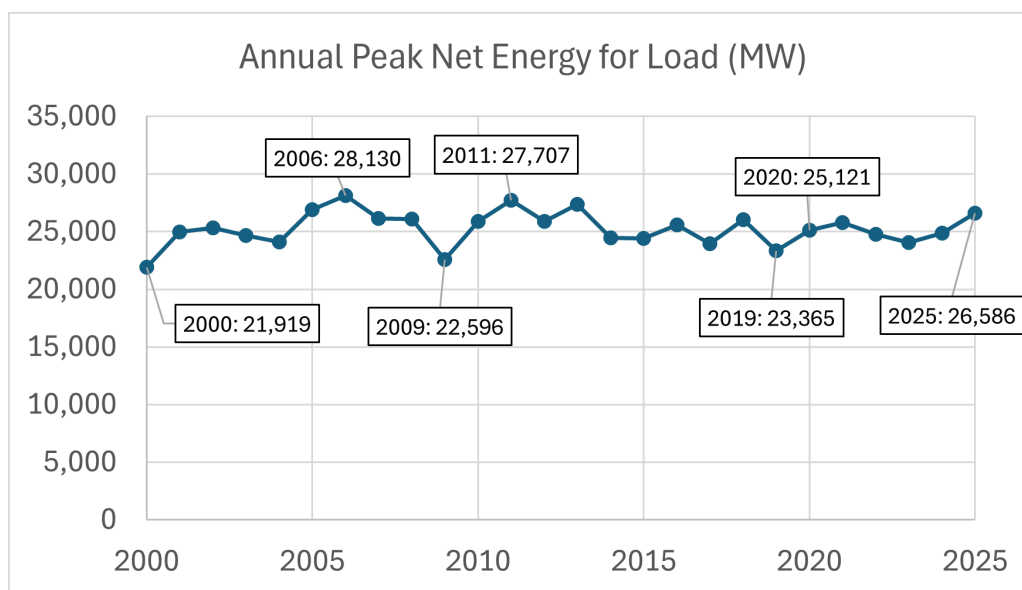


Figure 7: Annual Peak Net Energy for Load (*ISO New England, n.d.-a*)

Peak demand is met through a combination of generation sources. Nuclear provides a steady baseload output. Dispatchable generation, particularly natural gas, ramps up as needed to meet demand. In addition to these major utility-scale plants, there are approximately 100 plants designated for peak demand response. These “peaker plants” across New England only operate 2-7% of the time to meet demands on peak load days. Most of these plants are over 30 years old and run off natural gas or oil.

## Plan for ISO-NE to Transform to an Electricity-Based Society by 2050

### Baseline ISO-NE System

The previous section outlined the baseline condition of ISO-NE and described the starting point for the pathway for an electricity-based society by 2050. It is important to understand these baseline conditions in order to create an electrification plan because the region is not starting from zero. The region already has large low/no carbon assets. However, natural gas still leads as the primary source of generation, meaning that the main barrier to electrification is finding a way to safely replace this using new generation, storage, imports, and demand response tactics.

### Expected Load Growth

The future of load growth is not just a matter of how much annual energy demand increases, but importantly relies on when demand peaks. Historically, New England has been a summer-peaking region due to the common use of air conditioning on hot days. On the hottest days, these summer peaks have averaged around 25,600 MW, and on the coldest days average around 21,000 MW (ISO-NE). However, these summer peaks have not been wildly growing over the past 20 years due to energy efficiency improvements and behind-the-meter solar generation as discussed in the previous section.

ISO-NE's forecasts show that winter peaks will begin to grow much faster than summer peaks as building heating and transportation become further electrified. Based on the summer/winter peaks described in the first section, it can be seen that the summer peak will grow ~6% from 2026 to 2035. The winter peak will grow ~29% over the same time period. By 2035, the summer and winter peaks will be nearly equal, as seen in Figure 8. A takeaway from this observation is that winter reliability will need to be considered heavily while attempting to transition to an electricity-based system.

The following assumptions were made to model the peak and overall demand projections:

#### **Summer Peak Projection**

- Growth remains modest due to efficiency and solar.
- Assume the same 6% growth as predicted by ISO-NE for 2026-2035 and apply for 2035-2050.
- Although overall demand will be increasing, we will assume the improvements will offset this difference.

### Winter Peak Projection

- Winter peaks will grow faster as heat pumps and EVs add cold-weather load.
- We do not believe the growth pattern of 29% will apply from 2035-2050.
  - The transition push will be more concentrated in earlier years (from 2026-2035).
  - Grid stability will (optimistically) increase and storage methods will be more efficient and widely used.
- Assume a winter peak growth rate of ~17% from 2035-2050 to balance transition efforts with a stabilizing electric society.
  - 17% was chosen as the balancing number between ISO-NE prediction of summer growth at 6% and winter growth at 29%.

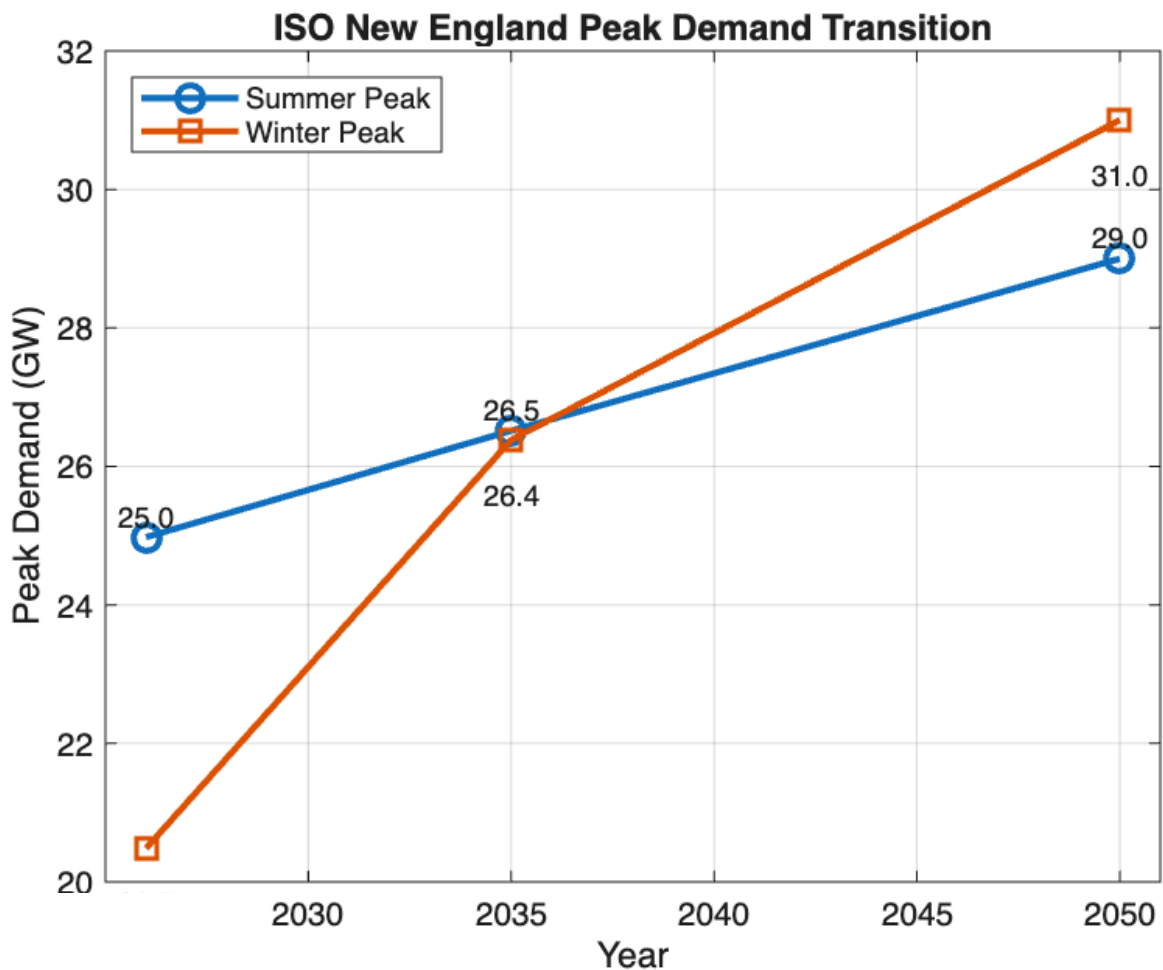


Figure 8: Peak Demand Transition



### Overall Energy Demand

- ISO-NE predicts a demand of 130,665 GWh by 2035.
- A logistic (S-curve) growth is used to predict demand through 2050.
  - Early growth (2025-2035): modest increase, estimate from ISO-NE used.
  - Mid-period (2035-2045): accelerated electrification from addition of EVs, heat pumps, etc.
  - Late period (2045-2050): efficient gains and technology maturity allow demand to mellow out. Demand saturates.

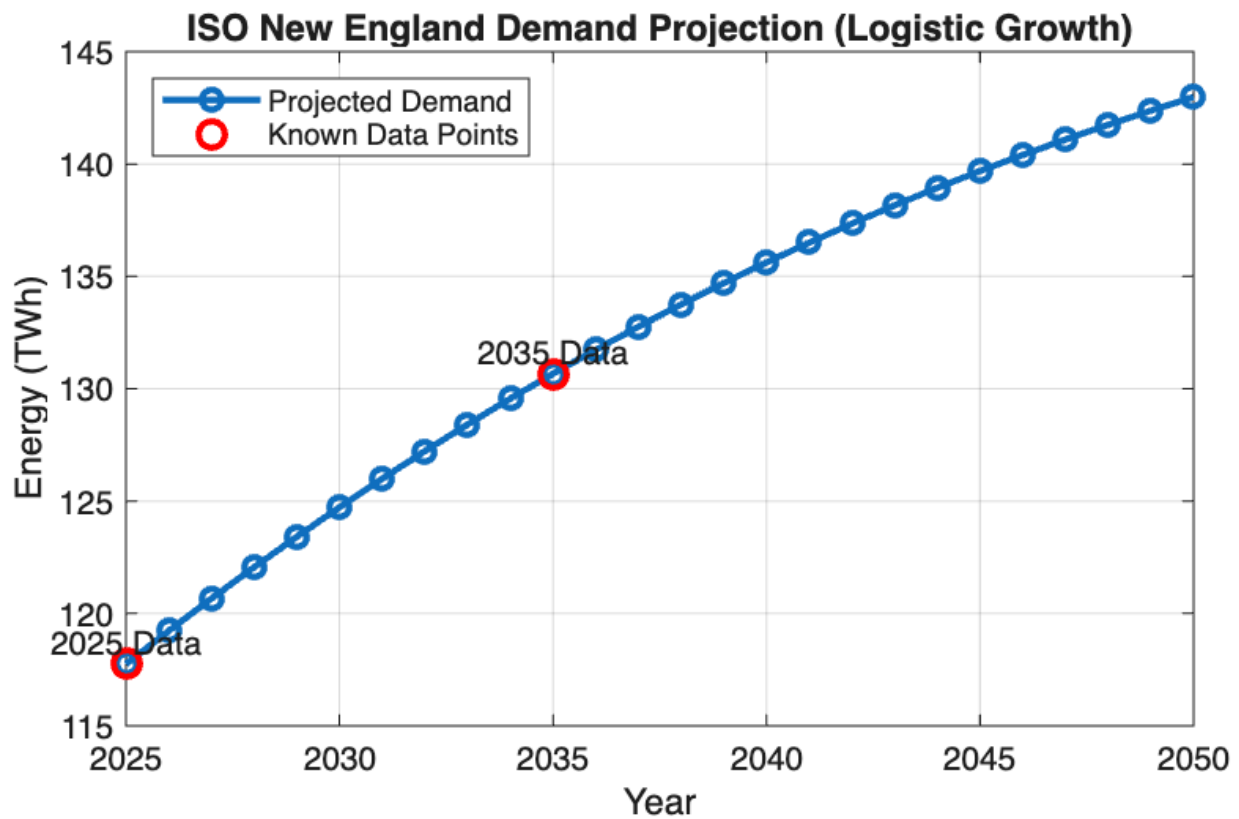


Figure 9: Demand Projection

| Metric               | 2025/26 Baseline | 2034/35 Projection     | 2050 Model Results |
|----------------------|------------------|------------------------|--------------------|
| Net energy for load  | 117,744 GWh      | 130,665 GWh            | 142,976 GWh        |
| Summer peak demand   | 24,978 MW        | 26,515 MW              | 29,000 MW          |
| Winter peak demand   | 20,483 MW        | 26,382 MW              | 31,000 MW          |
| Dominant peak season | Summer           | Nearly tied            | Winter             |
| Role of natural gas  | Primary resource | Declining contribution | Backup             |

Table 4: Demand and Peak Projections Summary

Table 4 (above) summarizes the inputs into the growth model and the results that are observed by 2050. The winter peak will exceed the summer peak by 2050, and overall demand rises to ~143,000 GWh. The next sections will talk about how generation will attempt to keep up with the rising demand and seasonal differences, and if this is even a realistic possibility for ISO-NE.

## Framework for Electrifying by 2050

The 2050 plan works like a balanced portfolio of energy resources that must sum to meet (or approach) electricity demand every hour of the year:

$$Load(t) \leq G_{firm}(t) + G_{wind}(t) + G_{solar}(t) + S_{discharge}(t) + I(t)$$

where G represents generation by type, S represents energy from storage, I represents net imports, and t is the hour of the year. Each resource has a purpose within the portfolio:

- **Firm (always available) low-carbon power:** nuclear and hydropower provide weather-independent, consistent generation. This is especially important during winter and nighttime when solar output is near zero.
- **Variable renewables:** offshore/onshore wind and solar provide a big share of annual clean energy, but their output varies by hour and season.
- **Daily balancing (short-term storage, demand response):** short-duration (2-8 hours) batteries charge during solar surplus and discharge during the late-afternoon and evening demand peaks.
- **Multi-day and seasonal backup:** long duration storage and hydro imports can cover any

extended gaps due to extreme weather conditions or excessive demand.

- **Transmission:** renewable generation is often far from cities, so transmission lines must carry power to these areas. Improved transmission is required for this plan to succeed.

This framework looks at energy from a few different perspectives: annually, seasonally, daily, and during peaks. These aspects must all be considered simultaneously to create a plan to electrify by 2050.

## Proposed 2050 Resource Strategy

### 1: Preserve Existing Firm Low-Carbon Resources

The plan assumes that the Millstone Nuclear Power Station and the Seabrook Station will continue operation through 2050. With their combined generation of 3.3 GW, they contribute to the firm power in the portfolio and are constant. The assumed capacity factor for these resources is 90%, so the overall annual output is about 26 TWh (*World Nuclear Association, n.d.*). This represents a significant share of the total generation, and is particularly valuable in events where variable renewables lag.

Existing hydropower plants, such as the Northfield Mountain pumped-storage facility, should be preserved and integrated into grid balancing efforts. Their flexibility makes them ideal for periods of peak demand rather than a baseload supplier.

### 2: Expand Offshore Wind as the Primary New Energy Source

New England has great potential for offshore wind; our plan has offshore wind providing the largest share of incremental clean generation. Currently, the ISO-NE interconnection queue is overflowing with wind and battery proposals. Most offshore proposals are located off the coasts of Massachusetts through Maine. Given these intentions, we assumed a target of 15-20 GW of offshore wind capacity by 2050.

Wind generation at a given hour is approximated as:

$$G_{wind}(t) = P_{wind} \cdot CF_{wind}(t)$$

At an average capacity factor of 0.44-0.50, and a winter capacity factor of 0.54, the intended capacity would result in 59-77 TWh per year in generation (*Gallucci, 2026*). Offshore wind is critical during the winter, since the capacity factor is stronger. Our prediction that winter peaks will exceed summer peaks by 2050 supports the efforts to get more wind capacity installed.

### 3: Continue Solar Growth as a Daytime Resource

Solar capacity should continue to expand. Our 2050 plan intends for more behind-the-meter installations that help lower midday demand that shows up on the grid.

Solar generation is described as:

$$G_{solar}(t) = P_{solar} \cdot CF_{solar}(t)$$

where  $CF_{solar}(t)$  peaks near solar noon and approaches zero at night. In New England, there is an approximate average solar capacity factor of 0.15 (*IECG, 2019*). However, this is an average that reflects the lows during the wintertime and highs during summer. Our plan targets 15-25 GW of solar capacity by 2050 which would result in 20-33 TWh per year of generation.

Although solar is a great resource, especially at the homeowner level, our plan does not emphasize it as the primary source of electrification for a few reasons. First, solar is highly variable based on season and time-of-day. It can't contribute much during the evening or winter. Second, excess solar would highly suppress midday grid but cause an equally dramatic effect on the evening ramp when solar resources aren't helping. So, the demand curve would be more steep. Because of this, our plan intends solar to function as an energy resource to offset some fossil generation, but it must be paired with strong storage in order to be reliable.

### 4: Storage in Two Layers

ISO-NE's own plan for electrification gives a warning that a future system built around wind, solar, and short-duration batteries alone would require ~4x today's total capacity, and that long-duration and seasonal storage would be critical as variable generation becomes more prominent. Storage is a key part of our plan, and we propose two layers:

#### Layer 1: Short duration batteries

- 2-8 hours of storage, target of 10-15 GW.
- The main focus of these batteries is to store the midday solar surplus and utilize it during the evening demand ramp.
- These batteries are meant to be fast-responding and address diurnal variability.

#### Layer 2: Long duration batteries and seasonal storage

- Hold energy across days or weeks, a target of 5-10 GW.
- Technologies include pumped hydro, flow batteries, compressed air, and thermal storage.
- These are meant to resolve any multi-day winter storms when wind or solar can't keep up.
- Excess energy will likely be stored in the spring and fall during low-demand periods and discharged during any high-demand events as needed.

- Pumped-hydro will mainly be used for grid-scale storage.

## 5: Maintain and Expand Imports and Transmission

Imports make up a large portion of the ISO-NE energy supply, about 8-10%. Much of the imports already come from clean sources, such as the hydropower from Québec. Imports are valuable because they can treat situations where there is physically not enough generation in the region. Our 2050 plan treats these imports as a permanent component of the portfolio, as we don't see enough clean capacity being built within this timeline. We target 5-8 GW of import capacity.

Efforts to improve transmission are equally important as the generation itself. The energy is essentially useless if it can't get to the area it's needed quickly and reliably. Renewable generation is concentrated in northern New England, while demand sits in metropolitan areas such as Boston and the Connecticut River Valley. Without expanded transmission in these demand centers, the generated power may become congested. These improvements are mainly a financial problem; investment and innovation are key to pushing this plan.

## Diurnal Variability and the Duck Curve

A successful transition to an electricity-based ISO-NE by 2050 needs to consider how demand changes throughout the day, not just over the year. System reliability depends not only on annual energy averages, but also on matching supply and demand at every hour.

As mentioned earlier in the report, ISO-NE demand on a daily basis is lowest overnight, increases in the morning, and peaks in the late afternoon or early evening. The net load fluctuations create a “duck-curve,” which is already visible today and will become more pronounced as solar expands.

On a typical summer day in 2050 (see Figure 10), the system is made up of a firm low-carbon baseline of approximately 9 GW (nuclear, hydro, and base imports), supplemented by solar peaking at roughly 8-9 GW near midday, wind fluctuating around 6 GW, and approximately 3-4 GW of battery discharge during the evening peak hours (hours 18-21). Together, these resources approach the projected summer peak of ~29 GW, but do not fully cover it during evening hours. So, the residual gap is closed by demand response and additional imports. This highlights the main challenge: the system has enough energy over the day, but not always at the right time. The largest mismatch occurs in the evening when solar output drops to zero but demand remains high.

Key strategies for diurnal variability:

- **Short-duration battery storage (2-8 hours):** charge during excess midday solar production and discharge during the evening demand.
- **EV charging demand response:** move flexible EV demand away from evening peaks to overnight or midday hours when generation is readily available and inexpensive.
- **Hydro dispatch:** hydro and pumped storage provide fast, dispatchable energy to support evening ramps and morning load increases.
- **Imports:** interregional ties can supply power when New England's own solar is unavailable.

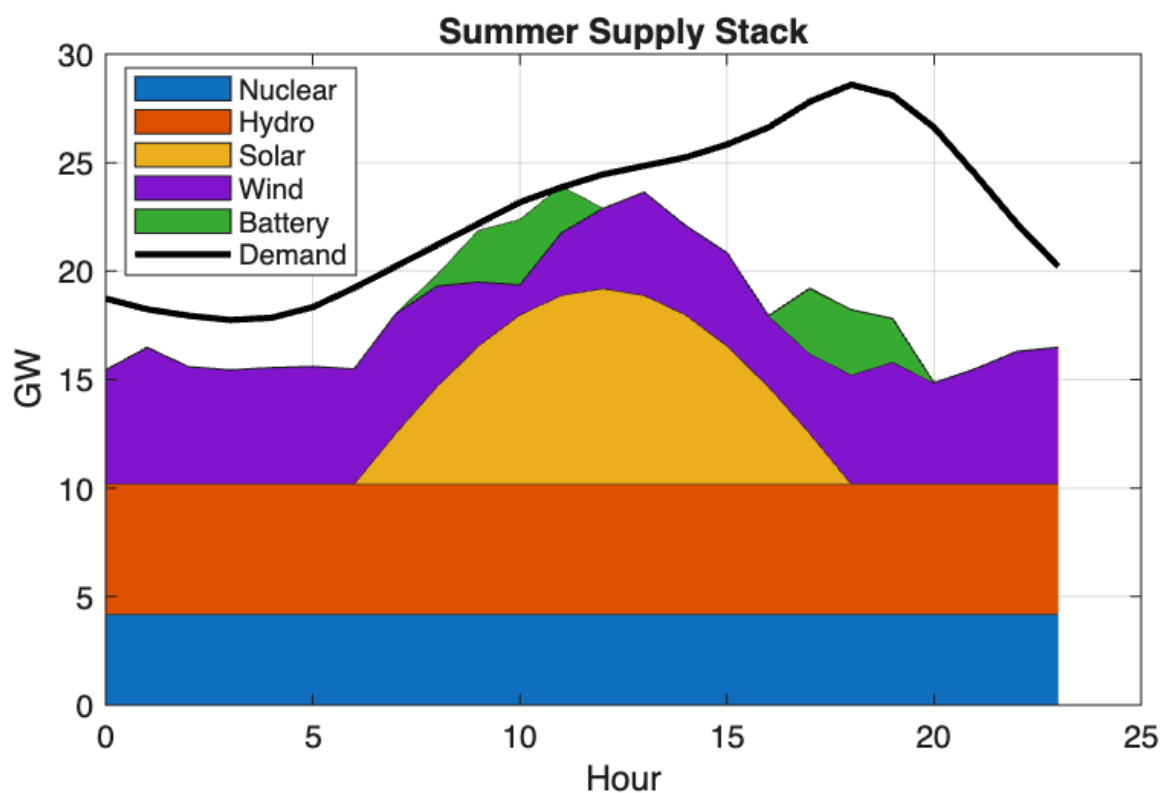


Figure 10: 2050 Summer Day - Supply Stack vs. Demand

## Seasonal Variability and Winter Adequacy

Seasonal variability is the most difficult reliability challenge for ISO-NE in a 2050 electrified system. As discussed earlier in the report, the system is transitioning from summer-peaking to winter-peaking. By 2050, winter peak demand is expected to reach ~31 GW,

exceeding the summer peak of ~29 GW. These projections are the principal design constraint of the 2050 resource portfolio.

Winter stress comes from three main factors:

- Reduced solar output due to shorter days and lower sun angle
- Wind variability during multi-day cold weather events
- Increased heating and transportation demand during cold periods

Unlike daily variability, these conditions can stretch from days to weeks and therefore cannot be addressed by diurnal storage technologies such as short-term batteries alone. They require resources with multi-day or multi-week energy capacity.

On a typical winter day in 2050 (see Figure 11):

- solar contributes only about ~1-3 GW at midday and none at night
- Firm low-carbon generation provides ~10 GW
- Offshore wind contributes ~8-10 GW on average

Even with these resources, the system still falls short of the winter peak of 31 GW. The remaining discrepancy must be met with long-duration storage and imports.

The figure also illustrates that, unlike on a summer day, there is little excess generation during the day. This means that batteries cannot rely on same-day charging. Instead, energy must be stored over longer periods or imported when needed.

Key strategies for seasonal variability:

1. **Firm low-carbon generation:** nuclear and hydro provide a constant output  $G_{firm}(t) \approx \text{constant}$ , independent of weather. These resources are important when wind and solar are weakest, and should be preserved for that reason.
2. **Offshore wind with strong winter capacity factor:** the ~0.54 winter capacity factor of offshore wind in the northeast makes it a critical resource for winter supply.
3. **Long-duration and seasonal storage:** pumped hydro, flow batteries, and thermal storage systems can store energy over days to weeks and discharge during longer winter deficits.
4. **Expanded Canadian hydro imports:** Québec's large reservoir hydro system can deliver consistent energy for weeks at a time, making it a reliable resource for New England.

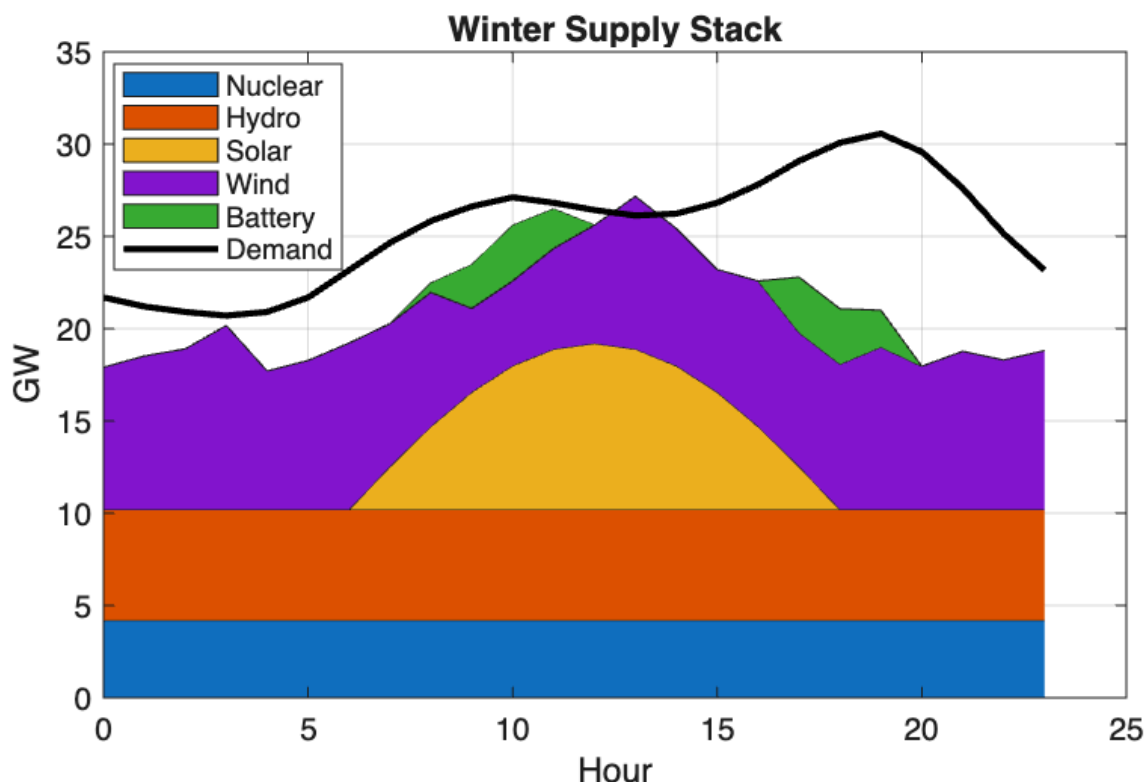


Figure 11: 2050 Winter Day - Supply Stack vs. Demand

## Meeting Peak Load

Meeting peak demand is a capacity problem, not just an energy problem. The system must have enough available and dispatchable resources during the highest-demand hours of the year.

Today, ISO-NE uses natural gas, nuclear, hydro, imports, operational reserves, and approximately 100 peaker plants. These peakers run only 2-7% of the time but account for about 15% of total installed capacity. They are used only when all other resources are fully utilized.

In a 2050 electrified system, this peak function shifts away from fossil peakers toward imports, storage (both short- and long-duration), demand response, and non-fossil firm resources. The generation-minus-demand chart shows the gap between clean generation and demand over time. This gap decreases steadily over time and approaches zero by 2050. This indicates that, in theory, the proposed resource portfolio is capable of meeting nearly all annual demand with low carbon generation.



However, it is important to recognize that this result represents an optimistic scenario. It assumes:

- The upper end of offshore wind deployment (closer to 20 GW)
- Strong solar buildout (toward 25 GW)
- Full implementation of both short and long term storage
- Expanded import capacity (5-8 GW) operating reliably when needed

Under these high-end assumptions, the system can eliminate the annual energy gap. In practice, delays in permitting, interconnection, or infrastructure buildout could reduce available capacity and reintroduce shortfalls.

Even in the optimistic case, peak reliability still depends on having sufficient dispatchable capacity during extreme conditions. A zero annual gap does not guarantee that all hourly peaks are covered, especially during winter.

The plan therefore does not require that natural gas be eliminated immediately. Instead, it should be reduced gradually. Gas plays a key role as a transition resource, specifically for winter reliability. The least efficient peaker plants should be retired first, while more flexible and reliable gas capacity remains available until storage and import capacity are proven sufficient enough to replace it.

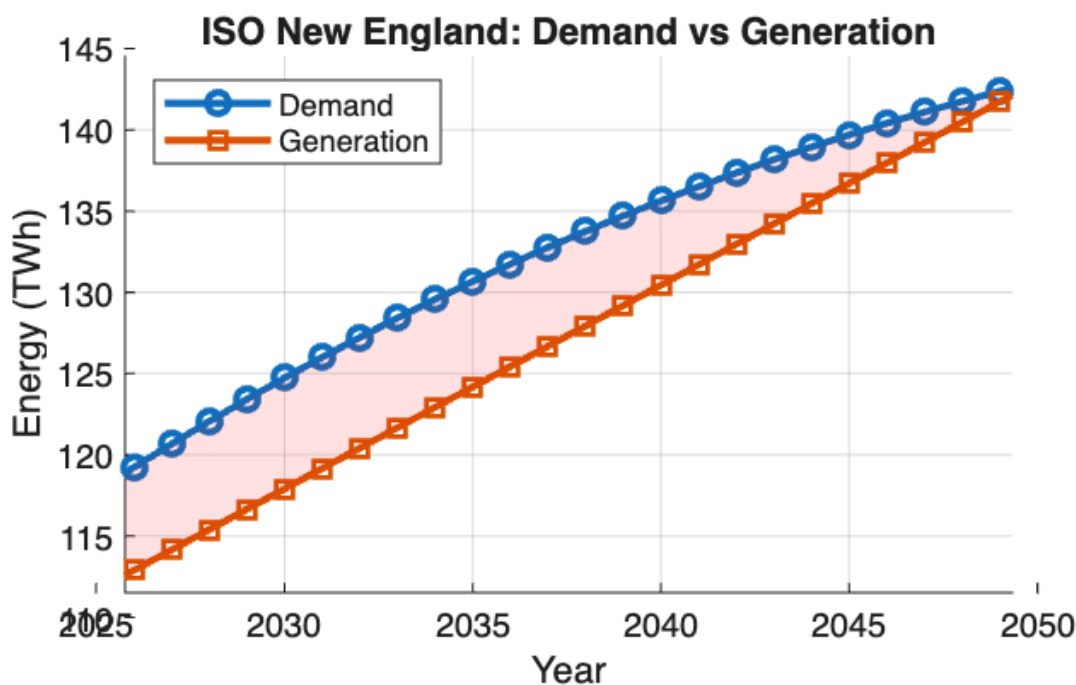


Figure 12: Demand vs. Generation from 2025 - 2050

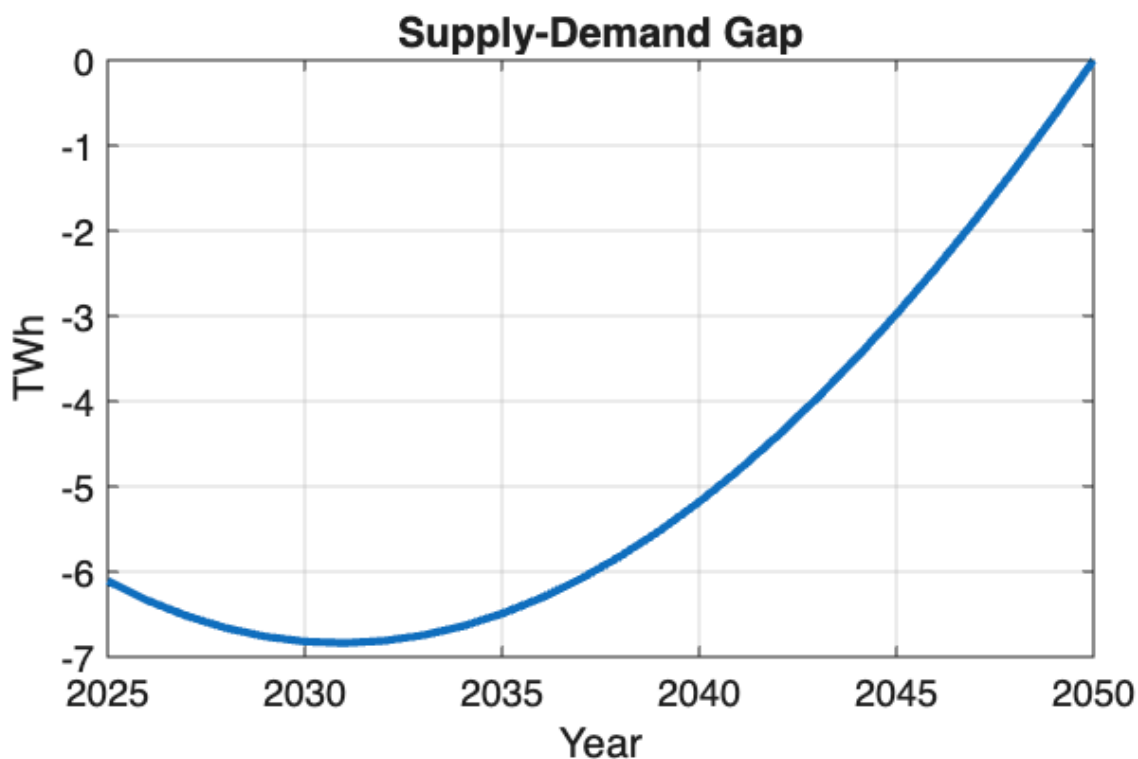


Figure 12: Energy Surplus/Deficit from 2025 - 2050

## Quantitative Resource and Planning Assumptions

Tables 5 and 6 summarize the assumptions used in the 2050 scenario. These values represent planning ranges rather than exact forecasts. Actual outcomes will depend on technology costs, permitting timelines, interconnection constraints, and policy decisions.

The results shown in the previous section, where the annual energy gap approaches zero by 2050, reflect an optimistic case that assumes deployment near the upper end of these capacity ranges. Lower buildout levels would result in a larger residual gap and greater reliance on imports or backup generation.

| Resource                         | 2025 Capacity | 2050 Target | Est. Annual Energy*      | Function                             |
|----------------------------------|---------------|-------------|--------------------------|--------------------------------------|
| Nuclear                          | ~3.3 GW       | 3–4 GW      | 23–31 TWh (CF ~0.90)     | Firm baseload                        |
| Offshore wind                    | ~0 GW         | 15–20 GW    | 59–77 TWh (CF ~0.44)     | Dominant new energy source           |
| Solar (utility + BTM)            | ~5 GW         | 15–25 GW    | 20–33 TWh (CF ~0.15)     | Daytime/diurnal resource             |
| Short-duration batteries         | ~1 GW         | 10–15 GW    | N/A (2–8 hr dispatch)    | Daily balancing & ramp support       |
| Long-duration / seasonal storage | ~0 GW         | 5–10 GW     | N/A (multi-day dispatch) | Winter adequacy & prolonged deficits |
| Hydro (incl. pumped)             | ~1.4 GW       | ~1.4 GW     | 7–10 TWh (retained)      | Flexible dispatchable & storage      |
| Imports (Canadian hydro, NY)     | 3–5 GW        | 5–8 GW      | Variable / on-demand     | Seasonal & peak reliability          |

\* Annual energy estimated using  $E = P \times CF \times 8,760 \text{ hr/yr}$ . Capacity factors: nuclear ~0.90; offshore wind ~0.44–0.50 (winter CF ~0.54 per Revolution Wind data); solar ~0.15 (New England); hydro ~0.55.

Table 5: Proposed 2050 Resource Portfolio

| Assumption                    | Value / Basis                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2050 annual demand            | ~143 TWh, driven by heating and transportation electrification                                                                                        |
| 2050 winter peak              | ~31 GW, exceeds summer peak; consistent with ISO-NE trend of winter peak growing faster than summer peak                                              |
| 2050 summer peak              | ~29 GW, modest growth from 25 GW (2026) due to continued AC demand and EV load                                                                        |
| Offshore wind capacity factor | 0.44–0.50 annual average; ~0.54 in winter                                                                                                             |
| Solar capacity factor         | ~0.15 for New England; near-zero generation in evening and overnight hours                                                                            |
| Nuclear operating assumption  | Millstone and Seabrook retained through 2050 if technically and economically feasible; ~3–4 GW at ~90% CF                                             |
| Import capacity               | Expanded to 5–8 GW via new Quebec hydro ties and NY interconnects; treated as firm seasonal resource                                                  |
| Natural gas role              | Declining from ~55% of generation (2025) to residual backup; retained as a transition and reliability resource, especially for winter peak conditions |
| Demand response               | Used for EV charging flexibility, building load shifting, and peak shaving; treated as ~1-2 GW of effective peak reduction                            |

Table 6: Key Planning Assumptions

Energy-to-capacity conversion for cross-check: the formula  $E = P \times CF \times 8,760 \text{ hr/yr}$  allows each capacity target to be translated into expected annual energy output.

Applied to the 2050 offshore wind target of 15–20 GW at  $CF = 0.44$ :  $E = 17.5 \text{ GW} \times 0.44 \times 8,760 = 67.4 \text{ TWh}$ .

Combined with nuclear (~27 TWh), solar (~26 TWh), and hydro (~9 TWh), the non-gas in-region generation total reaches approximately 125-130 TWh, which is close to the ~143 TWh demand target, with the remaining ~10-15 TWh covered by imports and retained flexible dispatchable capacity.

This balance assumes that the resource deployment occurs near the upper end of the proposed capacity ranges; lower deployment would increase the remaining gap.

The bar chart (Figure 13) illustrates the shift in generation mix. Note that the chart y-axis is scaled in units of  $\times 10^4$  GWh; the 2050 renewables bar at approximately  $9 \times 10^4$  GWh = 90,000 GWh (90 TWh) is consistent with the combined offshore wind and solar target described above.

Natural gas falls from approximately 60,000 GWh in 2025 to a residual backup role, while renewables increase by more than sixfold. This shift reflects a high buildout scenario and may vary depending on deployment timelines.

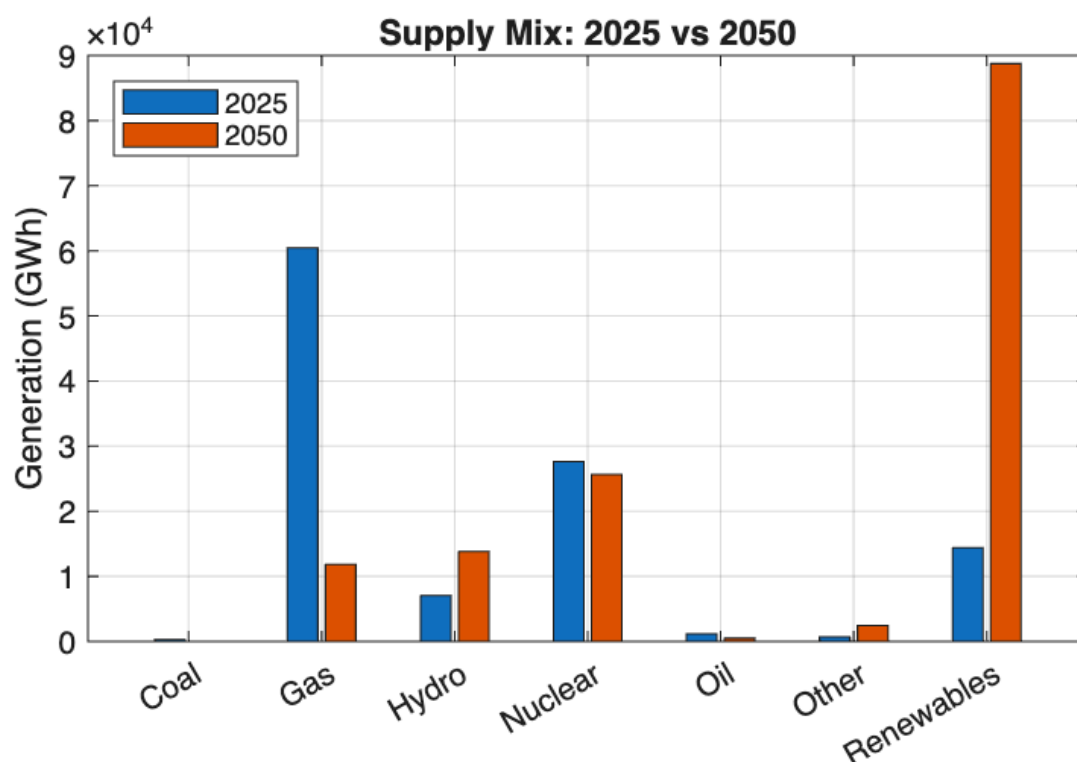


Figure 13: Annual Supply Mix - 2025 vs 2050 Scenario

## Plan Takeaways

ISO New England can transition toward an electricity-based system by 2050, but only through a balanced portfolio of resources.

The region already has a strong foundation in nuclear, hydro, renewables, and Canadian hydropower imports. However, electrification of heating and transportation will increase annual demand to ~143 TWh and shift the system toward more difficult winter peak conditions, with peak demand reaching ~31 GW.

The lowest-risk 2050 strategy therefore combines five methods:

- preserve existing firm non-fossil generation (nuclear and hydro) as the reliability backbone
- expand offshore wind as the primary new energy source, exploiting its favorable winter capacity factor
- add solar as a strong annual energy resource supported by short-duration batteries to manage the daily duck curve
- build long-duration and seasonal storage for winter reliability
- expand Canadian hydro imports and transmission to support peak demand

Under optimistic assumptions, where these resources fall at the higher end of the proposed capacity ranges, the system can approach a near zero annual energy gap by 2050. However, this depends on successful execution of large-scale wind, storage, and transmission projects. Realistically, delays or lower deployment levels would require sustained reliance on imports and flexible backup generation.

The key takeaway is that New England's main challenge is not producing enough clean electricity over the year, it is delivering enough reliable, dispatchable clean power during winter and evening peak periods. Every resource investment decision should prioritize this constraint.

## Meeting Peak Electric Demand at Lowest Cost

### 1. Avoiding High Cost Generation

Peak electricity cost in ISO-NE is strongly influenced by the resources required during a small number of high-demand hours. In the current system, these hours are often served by oil-fired peaker plants and high-cost natural gas units. Although these resources operate infrequently, they remain necessary for reliability because the grid must meet demand in real time. This makes them costly both operationally and economically.

### Low Utilization and High Effective Cost

Peaker plants are expensive because their fixed costs are spread over very little annual energy production. Although they may represent a meaningful share of installed capacity, they typically operate at low capacity factors. This can be shown using:

$$E = P \cdot CF \cdot 8760$$

1 GW of peaker capacity operating at a capacity factor of  $CF = 0.05$  produces:

$$E = 1 \cdot 0.05 \cdot 8760 \approx 438 \text{ GWh/yr}$$

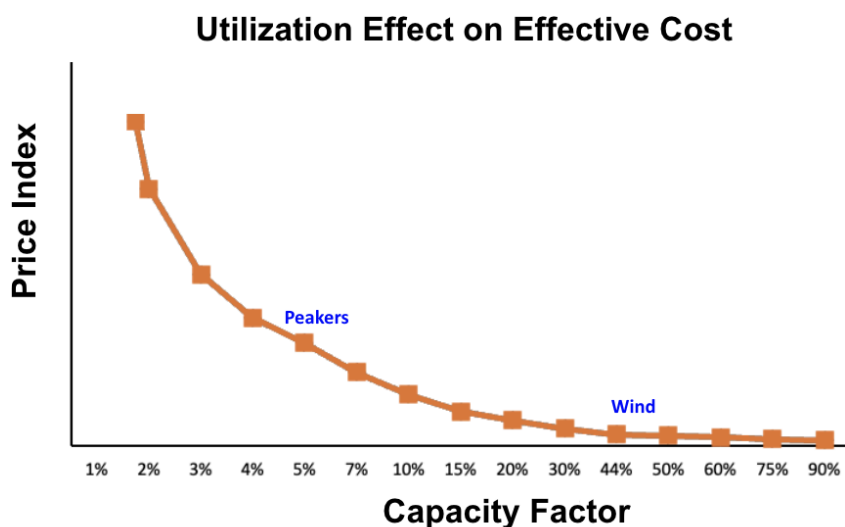
This is small compared with total system demand, approximately 143 TWh annually. Because the plant produces relatively little energy, its capital and fixed operating costs are spread over a small number of MWh. This increases the effective cost per unit of electricity.

This relationship is reflected in the levelized cost of energy:

$$LCOE = \frac{\text{Total Annual Cost (Capital + O\&M + fuel)}}{\text{Annual energy output}}$$

When annual energy output is low, the denominator is small, so the cost per MWh becomes high. This makes rarely used peak resources among the most expensive sources of electricity on a delivered-energy basis.

This utilization effect is shown in Figure 14. The figure isolates how lower capacity factors increase effective cost by spreading fixed costs over fewer units of electricity.



**Figure 14. Utilization effect on effective cost.**

Note: the peaker and wind labels are included as representative capacity-factor examples, not as full technology-specific LCOE comparisons.

## Marginal Pricing During Peak Hours

Electricity prices further amplify this issue because prices are set by the marginal generator, meaning the last and most expensive resource needed to meet demand:

$$C_{system}(t) = C_{marginal}(t)$$

Since electricity supply and demand must be balanced continuously, the system price reflects the cost of the next unit needed, not the average cost of all resources operating. During peak hours, the marginal unit is often a peaker plant or high-cost gas unit. As a result, even though these plants produce little energy annually, they can determine prices during the most expensive periods.

## Replacing Peakers with Flexible Resources

The proposed 2050 portfolio lowers peak cost by reducing reliance on these high-cost marginal resources. Short-duration battery storage, imports, and demand response each reduce



the need to dispatch expensive peakers during high-demand hours.

Short-duration battery storage, approximately 10-15 GW with 2-8 hours of duration, shifts energy from lower-cost periods to peak demand periods. In the modeled 2050 summer profile, batteries supply approximately 3-4 GW during evening peak hours. Each MWh discharged from storage replaces a higher-cost MWh that would otherwise come from a peaker or high-cost gas unit.

The cost benefit can be expressed as:

$$\Delta C = C_{peak} - C_{off\ peak}$$

- $C_{peak}$  = cost of high-cost marginal generation during peak hours
- $C_{off\ peak}$  = cost of lower-cost energy available during off-peak or surplus periods

Because  $C_{peak}$  is much larger than  $C_{off\ peak}$ , shifting energy into peak hours reduces operating cost.

### **Imports as Peak Generation Replacement**

Imports provide another way to avoid high-cost in-region generation. Expanding import transfer capability to 5-8 GW allows external hydroelectric resources to supply a portion of ISO-NE's peak demand. Québec's hydro system exceeds 40 GW of installed capacity and produces roughly 200 TWh annually, giving it the scale and flexibility to support exports during peak periods.

At a projected 2050 winter peak of approximately 31 GW, imports of 5-8 GW could supply about 16-26% of peak demand. This directly displaces high-cost marginal generation during peak hours. Here, the value of imports is mainly operational: they help meet peak demand without dispatching the most expensive local units.

### **Demand Response and Peak Dispatch Reduction**

Demand response further reduces the need for high-cost generation by lowering demand during critical hours. If demand response reduces peak demand by 1-2 GW, the effective peak

becomes:

$$Peak_{new} = Peak_{orig} - DR \approx 29 - 30 \text{ GW}$$

This directly reduces the amount of expensive generation that must be dispatched during peak periods.

## 2. Using Low Cost Energy More Effectively

Meeting peak electricity demand at lowest cost also depends on using low-cost energy when it is most valuable. Renewable resources such as solar and wind can provide large amounts of low-marginal-cost electricity, but their output does not always align with peak demand. As such, the system must not only generate cheap energy; it must also deliver that energy at the right time.

### Battery Storage and Peak Energy Shifting

In the proposed portfolio, substantial solar capacity is included because it provides very low-cost bulk energy. However, solar output is often highest during midday and lower during evening or winter peak periods. Without storage, some of this low-cost energy may be underutilized or curtailed.

Short-duration battery storage reduces this mismatch by capturing excess renewable generation and discharging it during peak demand. In the modeled system, approximately 3-4 GW of storage discharge occurs during peak hours, corresponding to roughly 10-15% of total peak demand.

This shows a meaningful portion of peak demand can be met using energy that was already generated during lower-cost periods, rather than by starting high-cost generators during the peak itself.

### Increasing Renewable Utilization

Storage increases the effective utilization of renewable generation. Without storage, excess energy during periods of high solar or wind output may be curtailed, meaning the system does not fully benefit from capacity that has already been built. By redirecting even part of this

energy to peak periods, storage increases the useful output of renewable resources.

In practical terms, more of the energy produced actually contributes to meeting demand. This lowers the cost per unit of useful electricity because the system gets more value from existing generation assets.

### **Flexible Demand and Load Timing**

Flexible demand provides a complementary way to improve energy utilization. Instead of only shifting electricity supply, the system can also shift some electricity demand. Discretionary loads such as electric vehicle charging, certain heating and cooling loads, and other flexible uses can be moved away from peak periods and toward hours with abundant renewable generation.

This reduces system cost in two ways. First, it lowers demand during high-cost hours. Second, it increases demand during periods when low-cost renewable energy is available, reducing curtailment and improving asset utilization.

Even modest adjustments in load timing can reduce reliance on the most expensive hours, especially when system prices are set by high-cost marginal resources.

### **Offshore Wind and Better Supply Timing**

Offshore wind also improves the timing of low-cost energy supply. While solar is strongest during midday and weaker during winter evenings, offshore wind can generate during evening and winter periods when demand is often higher and solar output is limited.

With projected annual output of 59-77 TWh, offshore wind contributes a large share of total electricity supply. Its value is not only that it adds energy, but that it adds energy during periods that better align with system needs. This reduces the amount of energy shifting required from batteries and flexible demand.

## Economic Value of Energy Shifting

The cost benefit of shifting low-cost energy into high-value hours can be represented as:

$$\text{Cost Savings} = E_{\text{shifted}} \cdot (C_{\text{peak}} - C_{\text{off peak}})$$

- $E_{\text{shifted}}$  = energy moved from low-cost to peak periods
- $C_{\text{off peak}}$  = low-cost renewable or surplus energy
- $C_{\text{peak}}$  = high-cost marginal generation

Even moderate amounts of shifted energy can produce meaningful cost savings because the difference between peak and off-peak costs can be large.

In this way, the portfolio lowers cost by improving the timing of both supply and demand. Storage shifts low-cost energy forward in time, flexible demand shifts some load toward low-cost periods, and offshore wind provides renewable generation that better matches winter and evening demand. These resources allow ISO-NE to use low-cost energy more effectively instead of relying only on additional peak generation.

## 3. Reducing Required Capacity and Infrastructure Buildout

The third cost pathway is reducing the amount of capacity the system must build and maintain. Because the electric grid must be sized to meet maximum demand, not average demand, even a small reduction in peak load can avoid major infrastructure costs. This includes avoided generation capacity, avoided transmission upgrades, reduced fixed O&M, and reduced need for rarely used reliability resources.

### Peak Demand Drives System Buildout

The system must maintain enough available capacity to meet the highest expected load. For example, if peak demand is approximately 31 GW, ISO-NE must have enough generation, storage, imports, transmission, and reliability margin to serve that level of demand, even if typical demand is significantly lower.

Reducing peak demand by 1-2 GW lowers the required capacity to approximately 29-30

GW. This avoids building resources that would only be needed during rare extreme conditions.

### **Demand Response as a Low-Cost Capacity Resource**

Demand response is one of the most direct ways to reduce capacity requirements. By shifting or curtailing load during high-demand periods, demand response reduces the maximum level of demand the system must be built to serve. Unlike new generation or transmission, demand response uses flexibility in existing consumption patterns.

This peak reduction can come from improved building efficiency, modest heating and cooling adjustments, behind-the-meter solar, and flexible loads such as electric vehicle charging. In ISO-NE, demand-side resources already contribute measurable capacity reductions, and scaling these programs can reduce the need for additional supply-side infrastructure.

### **Imports Reduce In-Region Build Requirements**

Imports also reduce required infrastructure buildout. In Section 1, imports were discussed as a way to replace expensive marginal generation during peak hours. Here, their role is different: imports reduce the amount of in-region capacity ISO-NE must build and maintain.

With projected import capability of 5-8 GW, ISO-NE can rely on existing external resources during peak periods rather than constructing equivalent local capacity. This lowers in-region capital expenditures while still supporting reliability.

The cost savings come from avoiding new local generation, transmission, and associated fixed O&M that would otherwise be needed to serve peak demand entirely within the region.

### **Long-Duration Storage for Extended Peaks**

Long-duration storage complements demand response and imports by reducing the need to overbuild the system for multi-day high-demand events. These extended peaks are especially important in a highly electrified 2050 system, where winter heating demand may create sustained periods of high load.

Instead of building enough additional generation to cover every multi-day peak event, stored energy can be discharged during these periods. This reduces the need for redundant firm

capacity that would otherwise be used only during extreme conditions.

The combined effect can be represented as:

$$P_{build} = P_{peak} - (DR + I + S)$$

- $P_{build}$  = capacity that must be built within the system
- $P_{peak}$  = original peak demand
- $I$  = imports
- $S$  = storage contribution

Figure 15 illustrates this capacity-reduction effect using an example case based on midpoint values from the proposed 2050 resource ranges.

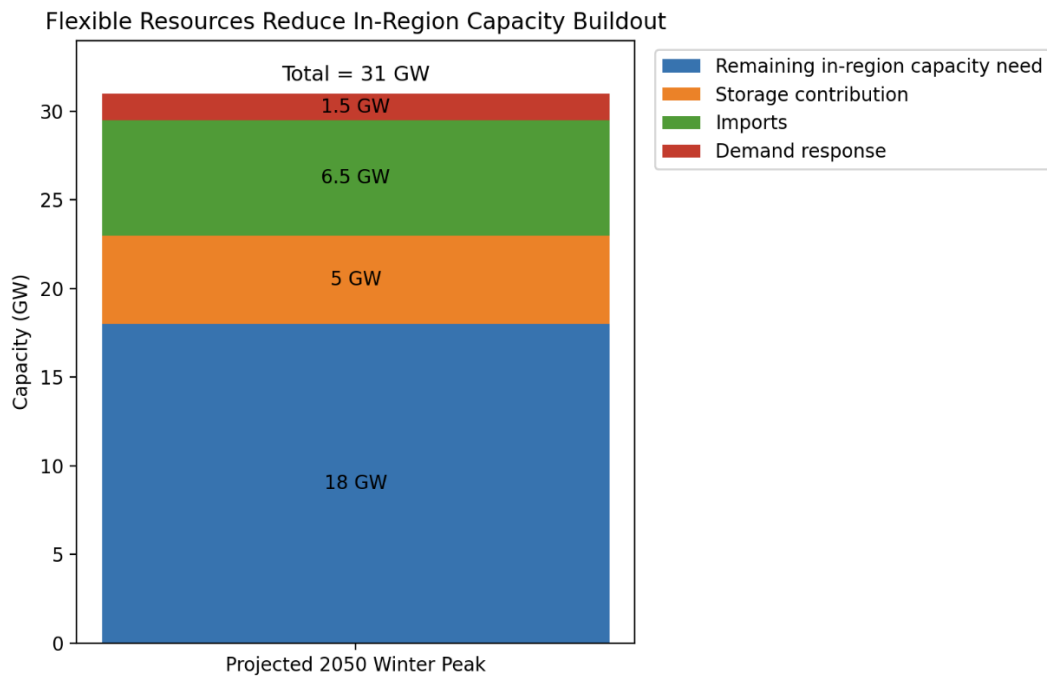


Figure 15. Flexible Resources Reduce In-Region Capacity Buildout

## Broader Infrastructure Savings

Reducing required capacity also lowers costs beyond generation. Lower peak demand reduces stress on transmission infrastructure, decreasing the need for upgrades and expansions

that would otherwise be required to deliver power during high-load conditions.

This creates savings through:

- avoided generation capacity additions
- avoided transmission upgrades
- reduced storage overbuild
- lower fixed O&M requirements
- reduced need for rarely used backup resources
- lower reliability-related buildout

Overall, minimizing required capacity reduces system cost by avoiding investment in infrastructure that would otherwise be used infrequently. Demand response lowers the peak itself, imports reduce the amount of capacity that must be built within ISO-NE, and long-duration storage helps cover extended peak events without requiring excessive firm capacity. This allows the system to meet reliability requirements with fewer total assets and lower fixed costs.

## Conclusions

### **Current System: Reliable but Inefficient Structure**

The current ISO-NE system maintains reliability through flexible natural gas generation comprising approximately 54% of supply and peaker plants that operate only 2-7% of the time. This allows real-time balancing, but it depends on capacity that is rarely used yet must always be available, meaning reliability is achieved through maintaining excess flexible capacity rather than efficiently utilizing energy. At the same time, demand patterns are shifting. The annual energy demand of about 117 TWh today remains manageable, but winter demand is increasing and approaching summer levels, creating more difficult operating conditions due to reduced solar output and potential fuel constraints. As a result, the current system's reliance on low-utilization flexibility becomes increasingly costly and harder to scale under electrification. Existing nuclear and hydropower resources also remain critical today, providing dependable low-carbon supply that reduces pressure on flexible fossil resources.

### **Energy Is Sufficient, Timing Is the Constraint**

Under optimistic buildout assumptions, the 2050 system demonstrates that total energy supply is achievable, with 125-130 TWh of in-region generation and 10-15 TWh of imports meeting the projected 143 TWh of demand. However, energy sufficiency does not ensure reliability, because generation is not aligned with demand. Solar produces large amounts of low-cost energy but is misaligned on a seasonal timescale with winter peak demand, while wind, although better aligned, remains variable. This leads to periods where energy is available but cannot be used when needed, meaning that system performance depends on delivering energy during peak hours rather than simply producing enough annually.

### **System Design: Coordinating Resources Across Time**

The proposed system coordinates resources across time by assigning clear and complementary roles to each resource. In our plan, offshore wind (15-20 GW, 59-77 TWh) provides the primary energy supply with strong winter contribution, while solar (15-25 GW, 20-33 TWh) lowers total system cost by supplying low-cost daytime energy. Storage and imports convert this energy into reliable supply: short-duration batteries (10-15 GW) shift energy daily



and provide 3-4 GW during average projected summer peak hours and margin for ultimate peak demand, while long-duration storage (5-10 GW) and imports (5-8 GW) ensure reliability during extended shortfalls. Existing nuclear and hydropower resources provide an essential reliability backbone throughout the transition. This system supports reliability through coordination across timescales rather than through overbuilding capacity, improving efficiency and lowering cost by aligning energy supply with when it is needed.

## **Peak Demand Drives Cost and System Design**

The cost analysis shows that system economics are dominated by peak conditions. In the current system, peakers account for around 15% of installed capacity but produce very little energy, leading to high effective costs and strong influence on peak pricing. Because these units set prices during peak hours, a small number of hours drives a large share of total system cost. The proposed system reduces this dependence by shifting low-cost energy to peak periods to replace the peakers. At the same time, the system must still be sized to meet the projected 31 GW winter peak, making peak demand the primary driver of infrastructure requirements. Reducing peak demand through 1-2 GW of demand response, along with storage and imports, directly lowers required capacity and avoids infrastructure that would otherwise operate only a small fraction of the time. Ultimately, the central challenge for ISO-NE is replacing low-utilization, high-cost peak capacity with flexible resources that can reliably meet winter and evening demand without overbuilding infrastructure, and we believe the proposed system is well-positioned to achieve this.

## Future Work

Several opportunities exist to further enhance performance, refine cost efficiency, and strengthen reliability under evolving system conditions.

### 1. Winter Reliability Under Extreme Conditions

The system is increasingly constrained by winter peak demand, making reliability during extreme conditions a critical uncertainty. While offshore wind, imports, and storage are expected to meet peak demand under typical conditions, performance during multi-day cold-weather events with low renewable output remains uncertain. Future work should evaluate system reliability under worst-case scenarios by modeling extended periods of high demand combined with low wind and solar generation. This would determine whether the assumed levels of long-duration storage and imports are sufficient or if additional firm capacity is required.

### 2. Best Storage Options

Storage plays a large role in enabling low carbon renewable resources, and as such we should analyze options more carefully. Storage options such as pumped hydro and various battery systems, as well as potential long-duration options such as hydrogen and compressed air, all differ significantly in cost, duration, efficiency, and geographic feasibility. The optimal choice depends on the required discharge duration, infrastructure constraints, and system integration considerations. Future work should evaluate these technologies in detail to determine which can most effectively provide reliability at the required storage duration, lowest cost, and scale within ISO-NE.

For example, further analysis could evaluate pumped hydro as a seasonal storage option by identifying viable sites in New England, estimating storage capacity and duration, and modeling dispatch over multi-day and seasonal timescales to determine its ability to mitigate winter peak deficits. This would be coupled with cost comparisons to alternative long-duration technologies and assessment of geographic, environmental, and transmission constraints on deployment. We would perform similar analysis for other options as well, balancing cost and feasibility constraints.

### 3. Numerical Cost Optimization of the Resource Portfolio

The proposed resource mix is based on engineering and economic reasoning but is not derived from a formal optimization framework. A more rigorous approach would involve solving a least-cost system optimization problem, minimizing total system cost subject to reliability constraints:

$$\min \sum (CAPEX \cdot CRF + O\&M + fuel)$$

with

$$Load(t) \leq Generation(t) + Storage(t) + Imports(t)$$

Additional constraints would be needed to capture storage state-of-charge dynamics, charging and discharging limits, transmission constraints, and resource availability profiles (e.g., wind and solar capacity factors as a function of time).

This approach would allow the levels of wind, solar, storage, imports, and demand response to be systematically optimized. It would also enable sensitivity analysis on key parameters such as fuel prices, technology costs, and demand growth, providing a more robust estimate of the lowest-cost system configuration. Such a model would likely take the form of a large-scale linear or mixed-integer optimization problem, requiring detailed input assumptions and computational methods to solve efficiently, but would provide a more precise and internally consistent determination of the optimal resource mix.

### 4. Demand Growth and Load Uncertainty

Future electricity demand is driven by uncertain factors such as electric vehicle adoption, electrified heating, and economic growth. The current analysis assumes a projected demand of approximately 143 TWh, but variations in these drivers could significantly affect both total energy needs and peak demand. Because major infrastructure such as offshore wind, transmission lines, storage, and import interconnections require long development and construction timelines, improving demand forecasts early is critical so needed capacity can be planned and built in advance. Future work should incorporate a range of demand scenarios to assess how sensitive the system design is to changes in load growth and peak timing.

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## Appendix A: Modeling Assumptions in Supply Stack Graphs

- Nuclear: baseload constant with no diurnal or seasonal variability. Output assumed to operate near a fixed level corresponding to ~90% capacity factor.
- Hydro: Modeled as a constant baseline contribution. Represents a combination of hydropower and firm imports.
- Solar: Modeled as daylight-only resource, using sinusoidal profile that peaks at solar noon and drops to zero at night. Winter peak output is reduced relative to summer.
- Wind: baseline + diurnal pattern + randomness (random vector used in Matlab code); real wind is unpredictable.
- Battery: Operates to reduce diurnal mismatch. Batteries charge during higher periods of solar generation and discharges when possible to meet peak demand, charges when solar is high
- Long-Duration/Seasonal storage: Included mainly in winter scenarios. Discharges during peak demand periods when other resources are unavailable.
- System Simplifications:
  - No transmission constraints are modeled
  - All resources are assumed to be available when needed within their modeled profiles
  - Results are optimistic, not a fully optimized dispatch model

## Appendix B: Matlab Model for Projections

The Matlab model will be uploaded through Box and include all code used to predict demand growth and generation graphs.